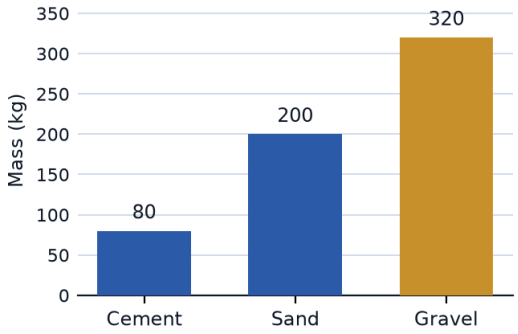
Problem-Solving, Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Multi-Part Ratios & Proportional Partitioning

In a concrete mix, the ratio of cement to sand to gravel by weight is $1 : 2.5 : 4$. How many kilograms of each material are needed to produce $600\,\text{kg}$ of concrete?



2 [Foundational] • [FRQ] • Target: Dimensional Analysis & Unit Conversion

Convert a speed of $90\,\text{km/h}$ into meters per second (m/s) and feet per second (ft/s, using $1\,\text{m} \approx 3.28\,\text{ft}$).

3 [Foundational] • [FRQ] • Target: Successive Percentage Changes

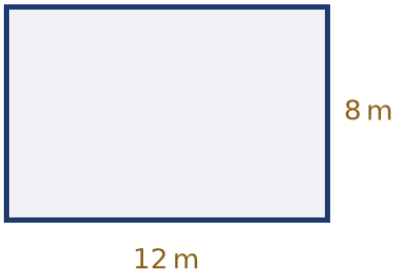
An electronics store marks up the wholesale cost of a laptop by 40% to set the retail list price, and then offers a 25% holiday discount off the list price. Find the net percentage profit over wholesale cost.

4 [Foundational] • [FRQ] • Target: Reverse Percentage Problems

After a 16% price reduction, a laboratory microscope sells for 630 dollars. What was the original price of the microscope before the reduction?

5 [Foundational] • [FRQ] • Target: Scale Drawings: Linear vs. Area Scaling

On an architectural blueprint drawn at a scale of $1\,\text{cm} : 2.5\,\text{m}$, a rectangular conference room measures $4.8\,\text{cm}$ by $3.2\,\text{cm}$. Find the actual perimeter (in meters) and actual floor area (in m^2) of the room.



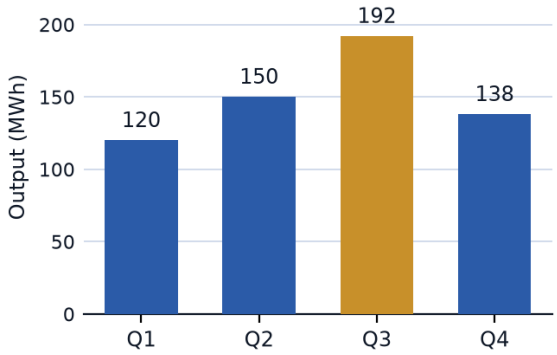
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6 [Foundational] • [FRQ] • Target: Combined Work Rates

Pump A can fill a reservoir in 6 hours, and Pump B can fill the same reservoir in 10 hours. If both pumps operate together, how many hours does it take to fill the reservoir?

7 [Foundational] • [FRQ] • Target: Reading Rates & Percentages from Bar Charts

The bar chart below shows quarterly solar energy output (in MWh) for a microgrid across four quarters. Calculate the percentage increase from Q1 to Q3 and the percentage of annual output generated in Q3.



Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

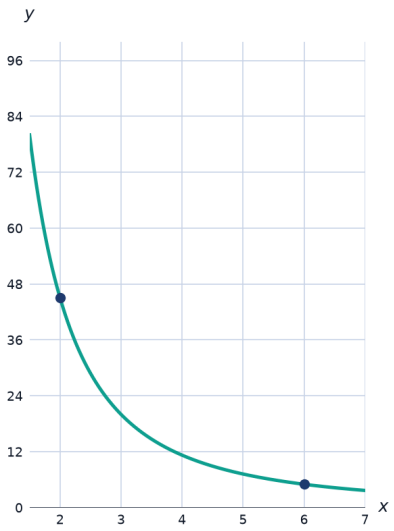
8 [Intermediate] • [FRQ] • Target: Concentration Enrichment Equations

A metallurgist has $80\,\text{g}$ of a 35% silver alloy. How many grams of pure (100%) silver must be melted into the alloy to raise the silver concentration to 60%?

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9 **[Intermediate]** • **[FRQ]** • **Target: Inverse-Square Variation & Percentage Change**

The illumination intensity I from a point light source varies inversely as the square of the distance d from the source ($I = k/d^2$). If $I = 180\,\text{lux}$ at $d = 2\,\text{m}$, find I at $d = 6\,\text{m}$, and determine by what percentage I changes when d increases by 25%.



10 **[Intermediate]** • **[FRQ]** • **Target: Weighted Average Speed Across Segments**

A commuter drives the first $60\,\text{km}$ of a trip at $80\,\text{km/h}$ and the remaining $90\,\text{km}$ at $60\,\text{km/h}$. Compute the total travel time and the overall average speed for the $150\,\text{km}$ trip.

11 **[Intermediate]** • **[FRQ]** • **Target: Percentage Inclusion-Exclusion**

In a student body of $1{,}200$ students, 65% take Spanish, 45% take French, and 20% take neither language. How many students take both Spanish and French, and what percentage of Spanish students also take French?

12 **[Intermediate]** • **[FRQ]** • **Target: Multi-Step Rate Unit Conversion**

Convert an irrigation flow rate of 450 gallons per hour into liters per minute, given $1\,\text{gallon} = 3.785\,\text{L}$. Round to two decimal places.

13 **[Intermediate]** • **[FRQ]** • **Target: Compound Percentage Recovery**

A stock portfolio drops by 20% in Year 1 and drops by another 10% in Year 2. By what percentage must the portfolio grow in Year 3 to return to its exact initial value?

14 **[Intermediate]** • **[FRQ]** • **Target: Chained Ratios & Homogeneous Expressions**

If $a : b = 3 : 4$ and $b : c = 6 : 5$, find the compound ratio $a : b : c$ in lowest integers and evaluate $\frac{2a + 3c}{4b}$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • [FRQ] • Target: Invariant Dry-Matter Percentage Problems • Cross-Lesson: Topic 1.01 (Conservation Linear Equations)

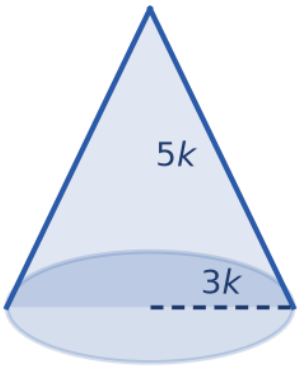
Fresh berries contain 80% water by weight, whereas dried berries contain 20% water by weight. How many kilograms of dried berries are obtained from $50\,\text{kg}$ of fresh berries?

16 [Challenging] • [FRQ] • Target: Power-Law Proportionality & Percentage Scaling • Cross-Lesson: Topic 4.01 (Fractional Exponents)

According to Kepler's Third Law, the square of a planet's orbital period T is directly proportional to the cube of its semi-major axis a ($T^2 = ka^3$). If Planet X has a semi-major axis that is 4 times that of Earth ($a_E = 1\,\text{AU}, T_E = 1\,\text{yr}$), find Planet X's orbital period and the percentage increase in T when a increases by 21%.

17 [Challenging] • [FRQ] • Target: Similarity Scaling of 2D Area & 3D Volume • Cross-Lesson: Topic 8.01 (3D Solids & Similarity)

Two similar right circular cones have surface areas in the ratio $9:25$. Find (a) the ratio of their heights, (b) the ratio of their volumes, and (c) the percentage by which the larger cone's volume exceeds the smaller cone's volume.



18 [Challenging] • [FRQ] • Target: Relative Risk, Sensitivity & Bayesian Proportions • Cross-Lesson: Topic 6.01 (Conditional Probability & Two-Way Tables)

A medical screening test for a condition present in 2% of a population has a sensitivity (true-positive rate) of 95% and a specificity (true-negative rate) of 90%. Out of $10{,}000$ people tested, how many test positive in total, and what percentage of those who test positive actually have the condition (Positive Predictive Value)?

19 [Challenging] • [FRQ] • Target: Repeated Percentage Decay & Exponential Inequalities • Cross-Lesson: Topic 4.01 (Exponential Decay) & Topic 2.01 (Inequalities)

Under continuous-rate radioactive decay, a sample loses 10% of its remaining mass every hour. Write an inequality for the number of hours t until less than 50% of the initial mass remains, and determine the smallest integer hour t that satisfies it.

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20 [Challenging] • [FRQ] • Target: Quadratic Revenue Elasticity from Percentage Changes • Cross-Lesson: Topic 3.01 (Quadratic Functions & Roots)

An airline increases its ticket price by $p\%$ and observes that ticket sales volume decreases by $\frac{p}{2}\%$. Express the percentage change in total revenue as a quadratic function of p, and find the non-zero value of p for which total revenue remains unchanged.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Simplifying Algebraic Ratios

If $\frac{3x}{5y} = \frac{9}{10}$, what is the ratio of x to y?

a. $2:3$

b. $3:2$

c. $9:50$

d. $27:50$

22 [Foundational] • [MCQ] • Target: Discount & Tax Calculations

A shirt originally priced at 60 dollars is on sale for 25% off. If an 8% sales tax is applied to the sale price, what is the final amount paid?

a. 45.00 dollars

b. 48.60 dollars

c. 49.80 dollars

d. 51.00 dollars

23 [Foundational] • [MCQ] • Target: Unit Rates & Time Conversion

A 3D printer produces 18 identical gears in 45 minutes. At this constant rate, how many gears can it produce in 2.5 hours?

a. 45

b. 54

c. 60

d. 75

24 [Foundational] • [MCQ] • Target: Equating Percentage Expressions

If 40% of x is equal to 16% of 150, what is the value of x?

a. 24

b. 48

c. 60

d. 96

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25 [Foundational] • [MCQ] • Target: Direct Proportions & Map Scales

A map uses a scale of 1 inch $= 15$ miles. If two cities are 4.6 inches apart on the map, what is the actual straight-line distance between them?

a. 60 miles

b. 64 miles

c. 69 miles

d. 75 miles

26 [Foundational] • [MCQ] • Target: Percentage Change Formula

The price of a commodity increased from 80 dollars to 100 dollars. What was the percentage increase?

a. 20%

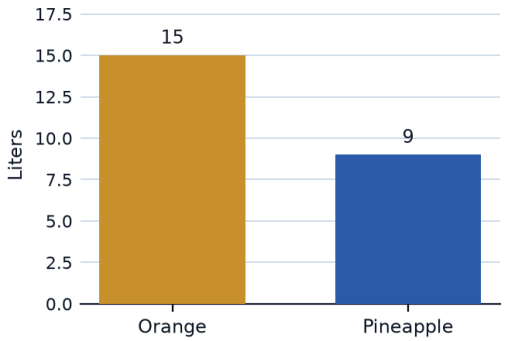
b. 25%

c. 80%

d. 125%

27 [Foundational] • [MCQ] • Target: Part-to-Whole Ratio Calculations

A fruit punch recipe calls for orange juice and pineapple juice in a ratio of $5 : 3$. How many liters of orange juice are in 24 liters of punch?



a. 9

b. 12

c. 15

d. 18

28 [Foundational] • [MCQ] • Target: Direct Variation

If y is directly proportional to x, and $y = 18$ when $x = 12$, what is the value of y when $x = 20$?

a. 24

b. 26

c. 30

d. 32

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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29 [Intermediate] • [MCQ] • Target: Compound Percentage Change in Area

The length of a rectangle is increased by 20% and its width is decreased by 15%. How does the area of the rectangle change?

a. It increases by 5%.

b. It increases by 2%.

c. It decreases by 2%.

d. It remains unchanged.

30 [Intermediate] • [MCQ] • Target: Inverse-Square Proportions

Sound intensity I is inversely proportional to the square of distance r from the speaker. If a listener moves from $r = 3\,\text{m}$ to $r = 6\,\text{m}$, the new intensity is what fraction of the original intensity?

a. $\frac{1}{16}$

b. $\frac{1}{4}$

c. $\frac{1}{3}$

d. $\frac{1}{2}$

31 [Intermediate] • [MCQ] • Target: Evaporation & Concentration Ratios

How many liters of water must be evaporated from $50\,\text{L}$ of a 12% salt solution to produce a 20% salt solution?

a. $15\,\text{L}$

b. $20\,\text{L}$

c. $25\,\text{L}$

d. $30\,\text{L}$

32 [Intermediate] • [MCQ] • Target: Algebraic Reverse Percentages

After a 20% discount, the sale price of a coat is p dollars. Which expression represents the original price of the coat before the discount?

a. $0.80p$

b. $1.20p$

c. $1.25p$

d. $\frac{p}{1.20}$

33 [Intermediate] • [MCQ] • Target: Updating Component Ratios

An alloy consists of copper and zinc in a $7:3$ ratio by mass. If $6\,\text{kg}$ of zinc is added to $40\,\text{kg}$ of this alloy, what is the new ratio of copper to zinc?

a. $7:4$

b. $14:9$

c. $7:5$

d. $2:1$

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34 [Intermediate] • [MCQ] • Target: Density, Mass & Volume Unit Conversions

Titanium has a density of $4.5\ \text{g/cm}^3$. What is the mass, in kilograms, of a solid titanium block with volume $2{,}000\ \text{cm}^3$?

a. $0.9\ \text{kg}$

b. $4.5\ \text{kg}$

c. $9.0\ \text{kg}$

d. $90\ \text{kg}$

35 [Intermediate] • [MCQ] • Target: Opposing Rates (Fill vs. Drain)

Pipe A can fill a tank in 4 hours, while a drain valve B can empty the full tank in 6 hours. If both Pipe A and drain valve B are accidentally left open when the tank is empty, how many hours will it take to fill the tank?

a. 8

b. 10

c. 12

d. 24

36 [Intermediate] • [MCQ] • Target: Chained Percentage Multipliers

If a is 25% greater than b, and b is 20% less than c, then a is what percent of c?

a. 95%

b. 100%

c. 105%

d. 110%

37 [Intermediate] • [MCQ] • Target: Harmonic Mean for Equal-Distance Round Trips

A cyclist rides uphill at $12\ \text{km/h}$ and immediately rides back down the exact same trail at $36\ \text{km/h}$. What is the cyclist's average speed for the entire round trip?

a. $18\ \text{km/h}$

b. $20\ \text{km/h}$

c. $24\ \text{km/h}$

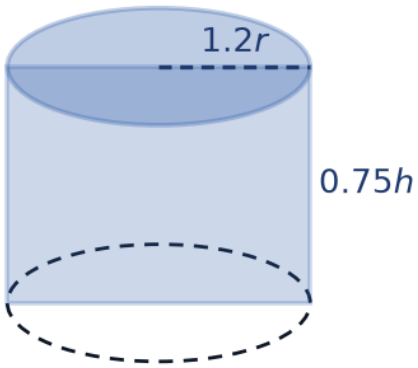
d. $27\ \text{km/h}$

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Multivariable Scaling of 3D Cylinder Volume • Cross-Lesson: Topic 8.01 (Volume of 3D Solids)

If the radius of a right circular cylinder is increased by 20% and its height is decreased by 25%, by what percentage does the volume of the cylinder change?



a. Decreases by 5%

b. Increases by 8%

c. Increases by 10%

d. Decreases by 10%

39 [Challenging] • [MCQ] • Target: Drain-and-Replace Mixture Modeling • Cross-Lesson: Topic 1.01 (Linear Conservation Equations)

A 10-liter radiator is filled with a 60% antifreeze solution. How many liters of the solution must be drained and replaced with pure (100%) antifreeze so that the resulting 10-liter mixture is 75% antifreeze?

a. $2.5\,\text{L}$

b. $3.25\,\text{L}$

c. $3.75\,\text{L}$

d. $4.0\,\text{L}$

40 [Challenging] • [MCQ] • Target: Compound Percentage Extrapolation • Cross-Lesson: Topic 4.01 (Exponential Growth)

An investment grows at a constant annual compound rate so that its value increases by 44% over 2 years. What is the percentage increase over 3 years?

a. 66%

b. 72.8%

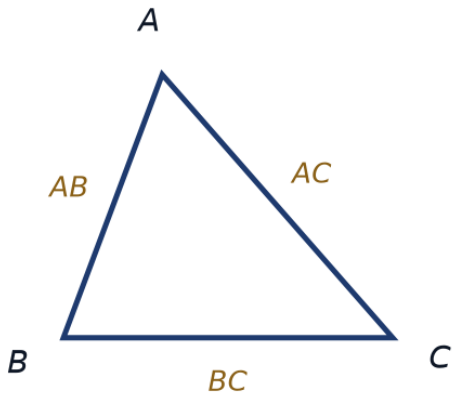
c. 88%

d. 95.2%

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41 [Challenging] • [MCQ] • Target: Area Ratios in Similar Triangles • Cross-Lesson: Topic 7.01 (Similar Triangles)

In ΔABC, point D lies on AB such that $AD : DB = 2 : 3$, and point E lies on AC such that $DE \parallel BC$. The area of ΔADE is what percent of the area of trapezoid $DBCE$?



a. 16%

b. 19.05%

c. 40%

d. 66.7%

42 [Challenging] • [MCQ] • Target: Symmetric Percentage Rise & Fall via Quadratics • Cross-Lesson: Topic 3.01 (Difference of Squares)

A store raises the price of an item by $x\%$ and later reduces the new price by $x\%$, resulting in a final price that is 9% less than the original price. What is the value of x?

a. 9

b. 15

c. 30

d. 45

43 [Challenging] • [MCQ] • Target: Ratio Rebalancing & Transfer • Cross-Lesson: Topic 2.01 (Linear Systems)

In a survey of 500 voters, the ratio of voters favoring Candidate A to Candidate B is $7 : 3$. How many voters must switch their support from Candidate A to Candidate B so that the new ratio becomes $1 : 1$?

a. 50

b. 75

c. 100

d. 125

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44 [Challenging] • [MCQ] • Target: Gear Ratios & Angular Velocity in Radians • Cross-Lesson: Topic 9.01 (Circles & Radian Measure)

Two gears are meshed together. Gear A has 36 teeth and rotates at 150 rpm. If Gear B has 24 teeth, through how many radians does Gear B rotate in 2 seconds?

a. 10π

b. 15π

c. 20π

d. 30π

45 [Challenging] • [MCQ] • Target: Linear Invariance of Equal Proportions • Cross-Lesson: Topic 10.01 (Rational Expressions)

To satisfy the synthesis condition, if $\frac{a}{b} = \frac{c}{d} = \frac{3}{4}$ with $b + 2d \neq 0$, what is the value of $\frac{a+2c}{b+2d}$?

a. $\frac{3}{8}$

b. $\frac{3}{4}$

c. $\frac{9}{16}$

d. $\frac{6}{7}$

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